On the Schrödinger Equation

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Abstract

The Schrödinger Equation is an indispensable postulate of quantum mechanics that governs the evolution of the wave function:

$$i\hbar\frac{\partial\psi(\vec{r},t)}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi(\vec{r},t) + \hat{V}(\vec{r},t)\psi(\vec{r},t).$$

Because it is taken as a postulate in quantum theory, one cannot derive the Schrödinger Equation in the "same manner" as other equations in Physics, such as, for instance, the Euler-Lagrange Equation.

This begs the question of *why* the Schrödinger Equation was chosen as a postulate, instead of some other relation, and this is what the paper aims to answer.

Errors found in this paper, queries, along with suggestions are most welcome.

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1 The Wave Packet

All wave functions in this paper are assumed to be of moderate decrease on the real line, and operators have not been explicitly indicated.

The wave equation is

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u.$$

We shall consider the one-dimensional case for simplicity; the solutions to this equation yield plane waves that propagate along the x-axis. The general form of a wave packet is the superposition of these solutions:

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{i(kx-\omega t)}\, dk. \tag{1}$$

Define the initial wave function $\psi_0(x) = \psi(x,0)$, so that

$$\psi_0(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi(k) e^{ikx}\, dx,$$

where, by the Fourier Inversion Formula,

$$\phi(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi_0(x) e^{-ikx}\, dx.$$

The de Broglie Relation implies the momentum is given by $p = \hbar k$, where k is the wave vector ($k = \vec{k}$ in $\mathbb{R}^1$). It follows that $dk = \frac{dp}{\hbar}$. Moreover, the Planck Relation yields the energy: $E = \hbar\omega$. Therefore, Equation 1 becomes

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \phi\left(\frac{p}{\hbar}\right) e^{i\left(\frac{p}{\hbar}x - \frac{E}{\hbar}t\right)} \,\frac{dp}{\hbar}.$$

Now, define $\tilde{\phi}(p) = \frac{1}{\sqrt{\hbar}}\phi(k) = \frac{1}{\sqrt{\hbar}}\phi\left(\frac{p}{\hbar}\right)$. Then, we may express the wave packet above in the form

$$\begin{aligned}
\psi(x,t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \sqrt{\hbar}\tilde{\phi}(p) e^{i\left(\frac{p}{\hbar}x - \frac{E}{\hbar}t\right)} \,\frac{dp}{\hbar} \\
&= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\phi}(p) e^{\frac{i}{\hbar}(px - Et)}\, dp.
\end{aligned}$$

2 The Schrödinger Equation

In Section 1, we showed that the equation of a wave packet in $\mathbb{R}^1$ is given by the equation[1]

[1]Recall that the kinetic energy is given by $T = \frac{p^2}{2m}$.

$$\psi(x,t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\phi}(p) \exp\left[\frac{i}{\hbar}\left(px - \left(\frac{p^2}{2m} + V\right) t\right)\right] dp;$$

It follows that

$$\begin{aligned}
i\hbar\frac{\partial}{\partial t}\psi(x,t) &= \frac{1}{\sqrt{2\pi\hbar}} \left(i\hbar\frac{\partial}{\partial t} \int_{-\infty}^{\infty} \tilde{\phi}(p) \exp\left[\frac{i}{\hbar}\left(px - \left(\frac{p^2}{2m} + V\right) t\right)\right] dp\right) && (2)\\
&= \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\phi}(p) \left(\frac{p^2}{2m} + V\right) \exp\left[\frac{i}{\hbar}\left(px - \left(\frac{p^2}{2m} + V\right) t\right)\right] dp && (3)\\
&= \left(\frac{p^2}{2m} + V\right) \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \tilde{\phi}(p) \exp\left[\frac{i}{\hbar}\left(px - \left(\frac{p^2}{2m} + V\right) t\right)\right] dp && (4)\\
&= \left(\frac{p^2}{2m} + V\right) \psi(x,t). && (5)
\end{aligned}$$

However, recall that the momentum operator in position space is given by[2]

$$\vec{p} = -i\hbar\vec{\nabla},$$

which is to say $p^2 = \hbar^2\nabla^2$, so in $\mathbb{R}^1$,

$$p^2 = \hbar^2 \frac{\partial^2}{\partial x^2}.$$

Substituting this into Equation 4,

$$\begin{aligned}
i\hbar\frac{\partial}{\partial t}\psi(x,t) &= \left(\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V\right)\psi(x,t)\\
&= \hat{H}\psi(x,t),
\end{aligned}$$

where $\hat{H}$ denotes the Hamiltonian. This is the (time-dependent) one-dimensional Schrödinger Equation.

References

[Zet22] Nouredine Zettili. *Quantum Mechanics*. Wiley, 2022. ISBN: 978-1-118-30789-2.

[2] See Equation 2.346 in [Zet22]